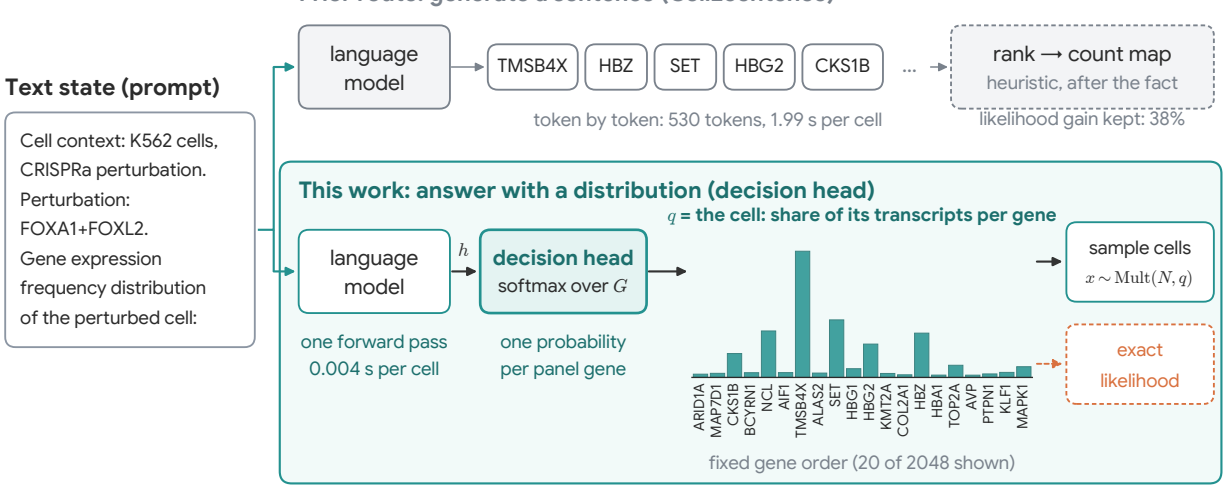
Prior route: generate a sentence (Cell2Sentence)
Text state (prompt)
Cell context: K562 cells, CRISPRa perturbation. Perturbation: FOXA1+FOXL2. Gene expression frequency distribution of the perturbed cell:
language model
TMSB4X
HBZ
SET
HBG2
CKS1B
...
rank → count map
heuristic, after the fact
token by token: 530 tokens, 1.99 s per cell
likelihood gain kept: 38%
This work: answer with a distribution (decision head)
q = the cell: share of its transcripts per gene
language model
h
decision head
softmax over G
sample cells
x ∼ Mult(N, q)
one forward pass
0.004 s per cell
one probability
per panel gene
exact
likelihood
ARID1A
MAP7D1
CKS1B
BCYRN1
NCL
AIF1
TMSB4X
ALAS2
SET
HBG1
HBG2
KMT2A
COL2A1
HBZ
HBA1
TOP2A
AVP
PTPN1
KLF1
MAPK1
fixed gene order (20 of 2048 shown)